ICT 1404 –Mathematics and Statistics for Computing

Probability Distributions

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Probability Distributions

A probability distribution is a statistical function that describes all the possible values and likelihoods that a random variable can take within a given range.

Types of Probability Distributions

There are many different classifications of probability distributions. The different probability distributions serve different purposes and represent different data generation processes.

- Binomial Probability Distributions
- Poison Probability Distributions
- Normal Probability Distributions
- ……………………..
- ……………………..

Binomial Distribution

The binomial distribution is a discrete distribution.

Charcteristics:

1. Each trial has only two outcomes.
One is said to be "Success" and other one is "Failure".

2. Number or trials are finite.

3. The probability of success for each trial is constant.

4. Trials are independent.

P(Success)= p   P(failure)=q   q=1-p

Binomial Distribution

If probability of occurring A is p, not occurring q and the number of trials is n. Then q=1-p
Then the probability of occurring A r times is,

$$p(r) = {}^{n}_{r}C p^{r} q^{n-r}$$

mean=np **variance= npq** **standard deviation=** $\sqrt{npq}$

${}^{n}_{r}C$ **=Number of different combinations of r items out of n items.**

$${}^{n}_{r}C = \frac{n!}{(n-r)!\,r!}$$

$${}^{n}_{r}C = \frac{\text{product of } r \text{ numbers from } n \text{ decreasing}}{\text{product of } r \text{ numbers from } 1 \text{ to } r}$$

Example1: ${}^{5}_{3}C$ $= \frac{5.4.3}{1.2.3}$ $= \underline{\underline{10}}$

$${}^{n}_{r}C = {}^{n}_{n-r}C$$

Example 2: ${}^{10}_{8}C$ $= {}^{10}_{10-8}C$ $= {}^{10}_{2}C$

$= \frac{10.9}{1.2} = \underline{\underline{45}}$

$${}^{n}_{0}C = {}^{n}_{n}C = 1$$

example:

It has found that the 10% of the transistors produced in a factory are defective. Find the probability that

(i) Exactly 2 transistors

(ii) At most 2 transistors

become defectives if 5 transistors are chosen randomly.

i.

N=5 r=2 p=0.1 q=1-0.1=0.9

$$p(x = r) = {}^{n}_{r}C p^{r} q^{n-r}$$

$$p(x = 2) = ({}^{5}_{2}C)(0.1)^{2}(0.9)^{5-2}$$

$$= \left(\frac{5x4}{1x2}\right)(0.1)^{2}(0.9)^{3}$$

$$= (10)(0.01)(0.729)$$

$$= \underline{\underline{0.0729}}$$

ii

N=5 r=2 p=0.1 q=1-0.1=0.9

$$p(x \le 2) = p(0) + p(1) + p(2)$$

$$= ({}^{5}_{0}C)(0.1)^{0}(0.9)^{5} + ({}^{5}_{1}C)(0.1)^{1}(0.9)^{4} + ({}^{5}_{2}C)(0.1)^{2}(0.9)^{3}$$

$$= 0.59049 + 0.32805 + .0729$$

$$= \underline{\underline{0.99144}}$$

Poisson Distribution

Founded by a French mathematician, Simeon Denis Poisson (1781-1840) . It is a discrete Probability distribution.

Applies with time intervals or space intervals.

Example:

- **Number of calls coming to phone in next 5 minutes.**
- **Number of road accidents occurred within one week.**
- **Number of misprints done per page by a typist.**
- **Number of customers arrive at an ATM in next minute.**

Charcteristics:

Probability is very low.(rare events)
Infinite no. of events. (not definite/uncountable)
Events are independent.
Time or space intervals are not intersected(overlap).

Poisson Distribution

If X is a discrete random variable where x=0,1,2,3,………… is distributed in a poison distribution, the probability when x=r is given by,

$$p(r) = \frac{e^{-\lambda}\lambda^{r}}{r!}$$

here λ= mean .

Poisson Distribution is a uniparametric distribution.

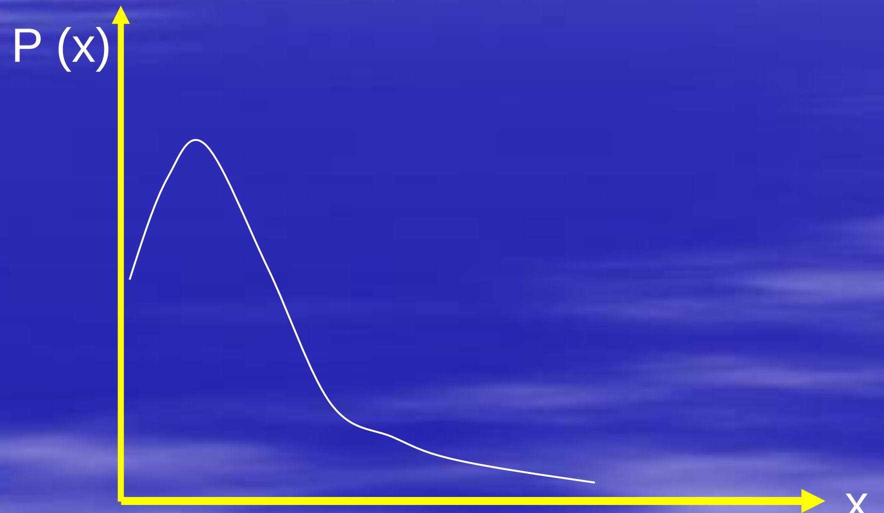
Mean = λ

Standard deviation= $\sqrt{\lambda}$

$e^{-\lambda}$

If a binomial distribution is with very low p and very higher n it can be used Poisson distribution. It is called Poisson Approximation to a Binomial distribution.

mean, $\lambda = np$.



Example 1

2% of items produced in a firm are founded to be defected and that number is in a Poisson distribution. If from a sample of 100 items one transistor is randomly selected, find the probability that (i) exactly one is defected (ii) No one is defected,

solution

(i) exactly one is defected

$$\lambda = np = 100x\frac{2}{100} = 2 \qquad r = 1, \lambda = 2$$

$$\mathrm{p(x = r)} = \frac{e^{-\lambda}\lambda^{r}}{r!}$$

$$\mathrm{p(x = 1)} = \frac{e^{-2}2^{1}}{1!} = 2e^{-2}$$

$$= 2(0.135)$$

$$= \underline{\underline{0.27}}$$

$e^{-\lambda}$

No one is defected,

ii. $$p(x=r)=\frac{e^{-\lambda}\lambda^{r}}{r!} \qquad r=1,\ \lambda=2$$

$$p(x=0)=\frac{e^{-2}2^{0}}{0!}=e^{-2}$$

$$=\underline{\underline{0.135}}$$

$e^{-\lambda}$

Example 2

The number of failures per week, of certain computer, has been found to follow a Poisson distribution with a mean 0.5. What is the probability that a computer has

(i) three or more failures in a given week ?

(ii) Exactly 3 failures in one month?

solution

(i)

$$p(x=r)=\frac{e^{-\lambda}\lambda^{r}}{r!} \qquad \lambda=0.5, \quad r=3,4,\ldots\ldots$$

$$P(x\geq 3)=P(x=3)+P(x=4)+\ldots\ldots$$

$$=1-[P(x=0)+P(x=1)+P(x=2)]$$

$$=1-[\frac{e^{-0.5}(0.5)^{0}}{0!}+\frac{e^{-0.5}(0.5)^{1}}{1!}+\frac{e^{-0.5}(0.5)^{2}}{2!}]$$

$$=1-e^{-0.5}[1+0.5+\frac{(0.5)^{2}}{2}]=1-0.607(1.625)$$

$$=\underline{\underline{0.013625}}$$

$e^{-\lambda}$

…solution

(ii) $\lambda = 0.5 \text{ x } 4 = 2, \quad r = 3$

$$\mathrm{p}(\mathrm{x} = \mathrm{r}) = \frac{e^{-\lambda}\lambda^{r}}{r!}$$

$$\mathrm{P}(3) = \frac{e^{-2}(2)^{3}}{3!}$$

$$= \underline{\underline{?}}$$

$e^{-\lambda}$

Normal Distribution

A most useful continuous probability distribution in statistics is the normal distribution. It has some properties that many situations in which it can be applied to make inferences.

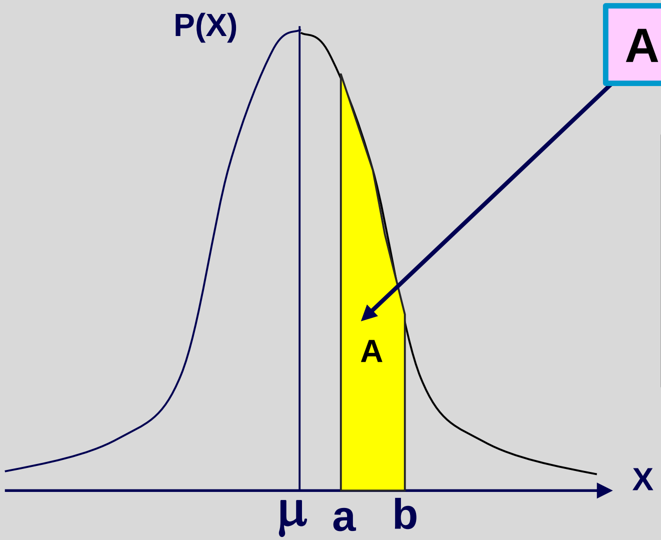
The normal distribution was first discovered by *De Moivre (1667-1754)* as a limiting form of the binomial distribution model in 1733. But later in 1809 *Karl Gauss* refer to this and this distribution is also known as Gaussian distribution.

Characteristics:

1. It is symmetric.
2. Mean, mode and median are same.
3. Bell shaped.
4. Curve does not meet x-axis.
5. Total of the area under the curve is 1.
6. The probability density function

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

If a random variable X is normally distributed The probability that X is between a and b, P(a<x<b) is given by the area under the curve of the standard normal distribution.

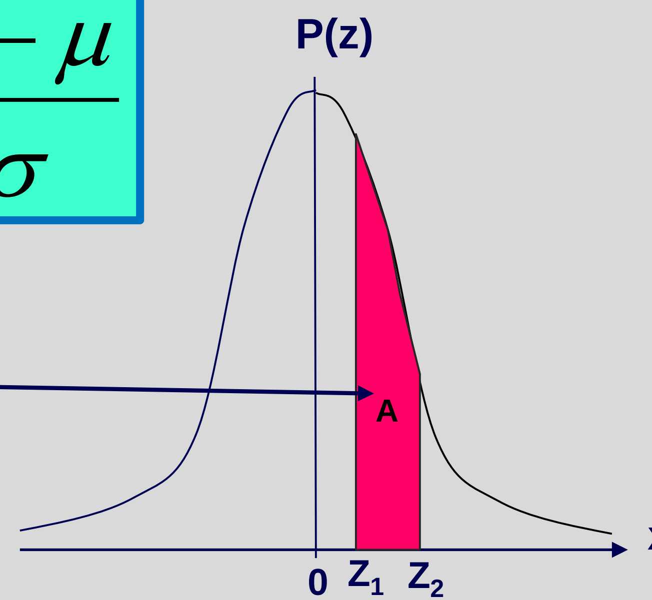


Area, A= P(a<x<b)

Any value of x is converted into a z value using following formula. This is called standardization.

$$z = \frac{x-\mu}{\sigma}$$

Area, A= P(Z_1< Z< Z_2)



You are provided tables to find this area.

standard normal distribution.

Example 1

If a random variable X is normally distributed with a mean of 10 and standard deviation 2 find the following probabilities.

(i) $P(x<7)$

(ii) $P(9<x<11)$

(iii) $P(11.5<x)$

(iv) $P(6.5<x<9.5)$

(i) P(x<7)

$\mu = 10$ $\sigma = 2$

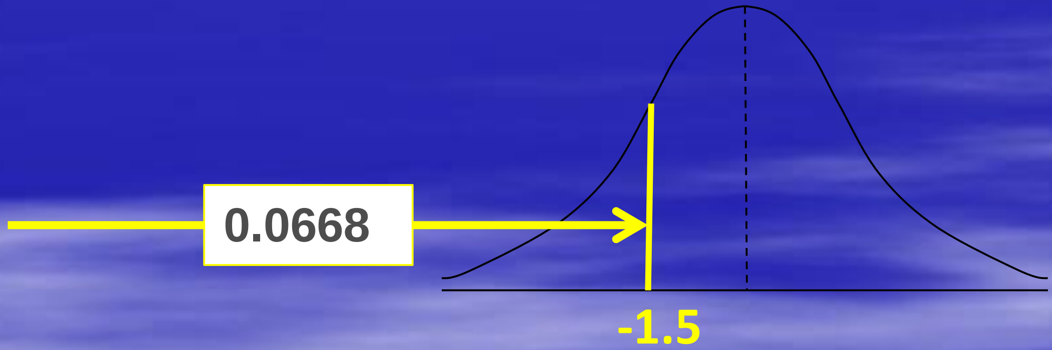
$$z = \frac{7-10}{2} = -1.5$$

When x= 7 then z= -1.5

$$z = \frac{x-\mu}{\sigma}$$

therefore P(x<7) = P(Z < -1.5)

Z table is used to find the area upto -1.5.



P(x<7) = 0.0668 = 6.68%

Z Table

solution (ii) $P(9<x<11)$

$\mu = 10$ $\sigma = 2$

$$z = \frac{x-\mu}{\sigma}$$

$$z = \frac{9-10}{2} = -0.5$$

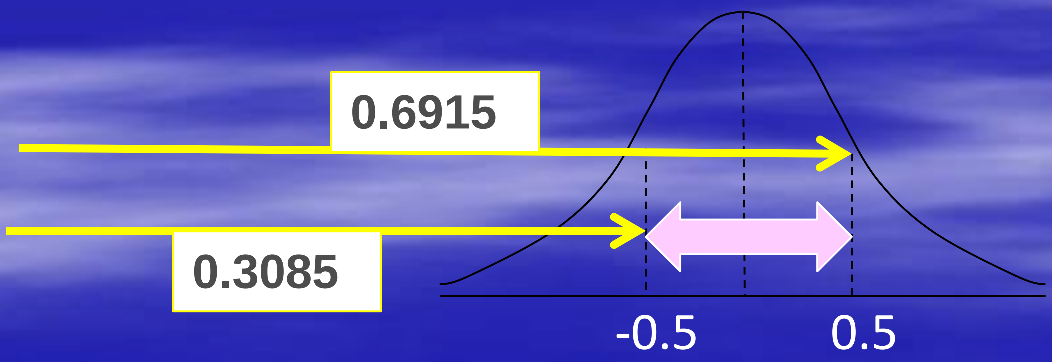
When x= 9 z= -0.5

$$z = \frac{11-10}{2} = 0.5$$

When x= 11 z= 0.5

$P(9<x<11) = P(-0.5 < Z < 0.5)$

To find area between -0.5 and 0.5 use -z table and +z table



$P(9<x<11) = 0.6915-0.3085=$ 0.383 = 38.3%

Z Table

(iii) P(11.5<x)

$\mu = 10$ $\sigma = 2$ $z = \dfrac{11.5 - 10}{2} = 0.75$

When x= 11.5 z= 0.75

$$z = \frac{x - \mu}{\sigma}$$

P(11.5<X) = P(0.75 < Z)

To find area upto Z=0.75 use +z table.

P(11.5<x) = 0.2234 = 22.34%

Z and t Table

solution (iv) $P(6.5<x<9.5)$

$\mu = 10$ $\sigma = 2$

$$z = \frac{x-\mu}{\sigma}$$

$$z = \frac{6.5-10}{2} = -1.75$$

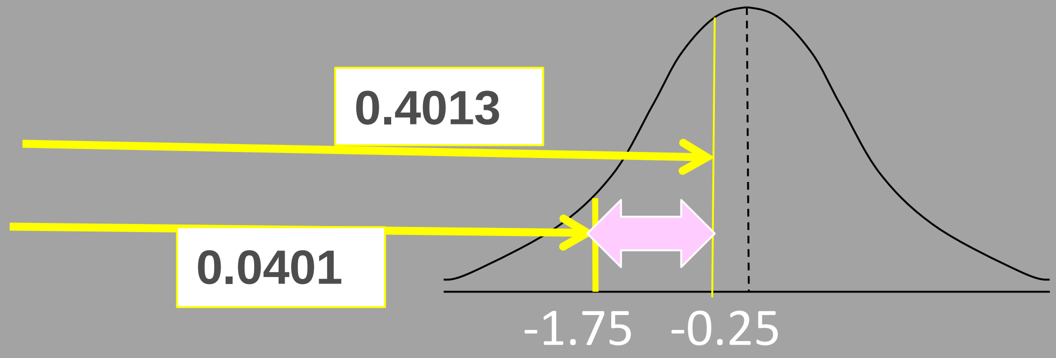
When x= 6.5 z= -1.75

$$z = \frac{9.5-10}{2} = -0.25$$

When x= 9.5 z= -0.25

$P(6.5<x<9.5) = P(-1.75 < Z < -0.25)$

To find area between -1.75 and -0.25 use -z table.



$P(6.5<x<9.5) = 0.4013-0.0401= 0.3612 = 36.12\%$

Z Table

Problem 2

The marks of a group of students has found normally distributed with mean 55 and standard deviation 10. Find the probability that a randomly selected student has obtained

i.A mark less than 75

Ii a mark between 40 and 50.

Find the lowest mark out of the 10% the students who scored highest marks.

Z Table

(i) P(75<x)

$\mu = 55$ $\sigma = 10$

$$z = \frac{x - \mu}{\sigma}$$

When x= 75 z= 2

P(75<X) = P(2 < Z)

P(75<x) = 1 - 0.9772 = 0.0228= <u>2.28%</u>

(ii) P(40<x<50)

When x= 40 z= -1.5

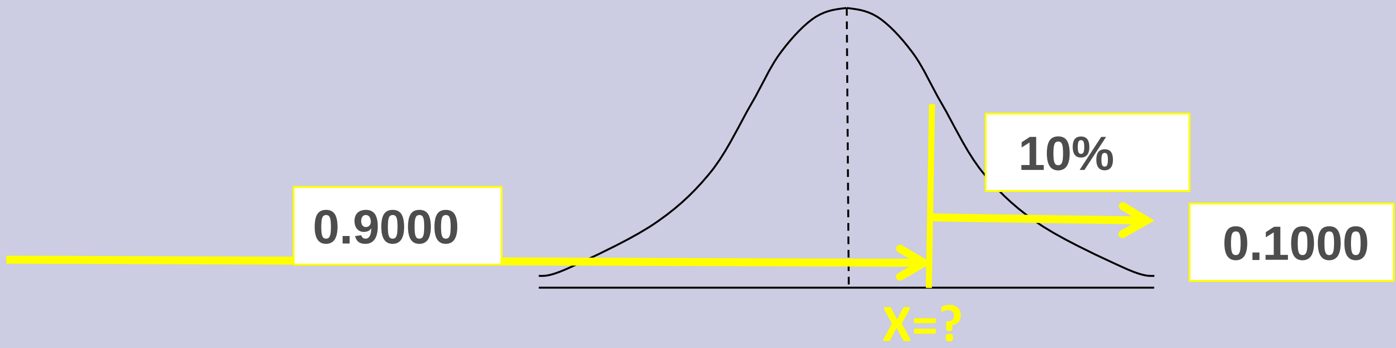
$$z = \frac{40 - 55}{10} = -1.5$$

When x= 50 z= -0.5

$$z = \frac{50 - 55}{10} = -0.5$$

P(40<x<50) = P(-1.5 < Z < -0.5) = 0.3085-0.0668 = <u>0.2417</u>

<u>Z Table</u>



Area above 10% is equal to area upto 0.9000 .

Z for area 0.9000, (from table) Z=1.28

$$z = \frac{x-\mu}{\sigma}$$

$$1.28 = \frac{x-55}{10}$$

$$x = 67.8$$

lowest mark of 10% the students who scored highest marks = 67.8

Z Table

Problem 3

Dennis Hogan Machinery Company has received a big order to produce electric motors for a manufacturing company. In order to fit in its bearing, the drive shaft of the motor must have a diameter of $5.1 \pm .05$ (inches) . The company's purchasing agent realizes that there is a large stock of steel rods in inventory with mean diameter of 5.07 inches, with a standard deviation .07 inches. What is the probability of a steel rod from inventory fitting the bearing ?

Problem 4

The income of a group of 10000 persons was found to be normally distributed with a mean, Rs. 7500.00 and standard deviation Rs. 500.00. Show that of this group of about 95% had income exceeding Rs. 6680.00 and only 5% had income exceeding Rs. 8320.00. What was the lowest income among the richest 100 persons?

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